

Course Code: CS1108 Name of the Programme: BCA (Hons.) Name of the Course: Linux and Shell Programming Semester: I		
Course Credit	No. of Hours Per Week (L+T+ P)	Total No. of Teaching Hours
4	3:0:2	75
Course Objectives:		
a) Provide a comprehensive understanding of the Linux operating system, its architecture, and core functionalities.		
b) Develop proficiency in using Linux command-line utilities, text-processing tools, and system operations.		
c) Enable students to write, debug, and execute shell scripts for task automation.		
d) Equip students with practical skills in system administration and routine Linux-based automation tasks.		

Course Outline (Syllabus of the Course)

Module - 1 : Introduction to Linux and the Command Line	No. of Hours
	9 Hrs.
History and Philosophy of Linux, Introduction to Open Source Software and Linux Distributions (e.g., Ubuntu, Fedora, CentOS), Linux System Architecture: Kernel, Shell, File System, Basics of Command Line Interface (CLI), Installing Linux (Ubuntu Server) on VirtualBox , Basic Navigation in Linux Terminal: cd, ls, pwd, mkdir, rm, cp, mv, Introduction to Text Editors: vi/vim	
Module - 2 : Core Linux Commands	No. of Hours
	9 Hrs.
Understanding the PATH Variable, Essential Commands: man, echo, printf, passwd, uname, who, date, File Handling Commands: cat, more, wc, lp, od, tar, gzip, File and Directory Management: pwd, cd, mkdir, rmdir, ls, cp, mv, rm, Security with File Permissions, Process and Disk Management: ps, w, ulimit, du, df, mount, umount, Networking Commands: arp, ftp, telnet, rlogin, Searching and Utilities: find, unlink, unmask	
Module - 3 : Text Processing and Advanced Utilities	No. of Hours
	9 Hrs.
Text Processing Tools: head, tail, sort, nl, uniq, grep (grep, egrep, fgrep), cut, paste, join, tee, File Comparison: comm, cmp, diff, Data Manipulation: tr, awk, sed, grep and sed: Usage, Scripts, Operations, Addresses, and Applications, Backup Utilities: cpio, tar, gzip	

Module - 4 : Shell Programming, Basics of Shell Scripting	No. of Hours
	9 Hrs.
Writing Shell Scripts, Using read and Command Line Arguments, Variables, Logical, and Conditional Operators (if, case), Computations with expr and String Handling, Loops: while, for, Advanced Features: set, shift, trap, and Debugging Shell Scripts, Input Validation and Data Entry Scripts, Introduction to Arrays and Functions in Shell	
Module - 5 : Advanced awk Scripting, Overview of awk	No. of Hours
	9 Hrs.
Structure, Execution, and Comments, Operators, Fields, and Built-in Variables, Output Formatting: print, printf, Escape Sequences, OFS Variable, User-Defined Variables and Parameter Passing, Control Structures: if-else, while, for Loops, Arrays: Declaration, Subscripts, Deleting Elements, User-Defined Functions and the getline Function, Regular Expressions and Advanced Use Cases	

Course Outcomes:
a) Understand the fundamental concepts and architecture of the Linux operating system.
b) Apply Linux commands to manage files, processes, and system operations effectively.
c) Develop and debug shell scripts to automate tasks within a Linux environment.
d) Analyze and process data using Linux text-processing tools such as grep , sed , and awk to solve real-world problems.

Text Books:	
1	Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., & Mackin, D. (2017). <i>UNIX and Linux system administration handbook</i> (5th ed.). Addison-Wesley Professional. ISBN: 978-0134277554
2	Das, S. (2012). <i>Unix concepts and applications</i> (4th ed.). McGraw Hill. ISBN: 978-0-07-063546-3

Reference Books:

1	Afzal, I. (2019). <i>A complete course on Linux bash shell scripting with real life examples</i> . Packt Publishing. ISBN: 978-1838984083
2	Rosen, K., Host, D., & Klee, R. (2007). <i>The complete reference UNIX</i> (2nd ed.). Tata McGraw-Hill. ISBN: 0-07-065836-6
3	Blum, R., & Bresnahan, C. (2015). <i>Linux command line and shell scripting bible</i> (3rd ed.). Wiley. ISBN: 978-1118983843

Reference Links

1	https://onlinecourses.swayam2.ac.in/aic20_sp05/preview
2	https://www.coursera.org/learn/hands-on-introduction-to-linux-commands-and-shell-scripting

Laboratory Components

1	Practice basic Linux navigation commands: cd, ls, pwd, mkdir, rm, cp, and mv.
2	Edit and save files using text editors like vi and vim.
3	Explore and understand the PATH variable and execute essential commands like man, echo, passwd, and uname.
4	Perform file handling tasks using commands like cat, more, wc, lp, od, and tar.
5	Compress and extract files using gzip, tar, and cpio.
6	Manage files and directories using commands such as pwd, cd, mkdir, rmdir, ls, cp, mv, and rm.
7	Monitor system processes and disk usage with ps, w, df, du, mount, and umount.
8	Use networking commands like arp, ftp, telnet, and rlogin to explore basic networking tasks.
9	Search for files and directories using commands like find, grep, and unmask.
10	Process text files using tools like head, tail, sort, nl, uniq, and cut.
11	Compare files using commands like comm, cmp, and diff, and manipulate data with tr, paste, and join.

12	Create and debug shell scripts that include variables, loops, and conditional statements (if, case, while, for).
13	Implement advanced shell scripting features like set, shift, trap, and arrays.
14	Write awk scripts to process and format data, exploring control structures (if-else, while, for) and arrays.
15	Use regular expressions with awk and sed to perform pattern matching and text processing.
16	Develop backup scripts using tar, cpio, and gzip to automate data archiving and recovery.

Lesson plan

Name of the Programme:	BCA(Hons.)	
Title of the Course:	Linux and Shell Programming	
Course code	CS1108	
Semester:	I	
Names of the Course Instructor(s)	Dr. K Thenmozhi	
Course credit	No. of hours per week (3+0+2)	Total no. of Teaching hours
04	05 Hours	75 Hours

Sessio n	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	1	History and Philosophy of Linux	Lecture, Discussion	Das Ch.1, Nemeth Ch.1
2		Open Source	Lecture, Q&A	Nemeth Ch.1, Swayam link
3		Linux Distribution	Lecture , Discussion	Das Ch.1, Nemeth Ch.1
4		Installing Ubuntu on VirtualBox	Guided Lab	Nemeth Ch.2
5		CLI Basics and Terminal	Hands-on Demo	Nemeth Ch.2, Coursera link
6		Linux System Architecture: Kernel	Lecture + Diagrams	Das Ch.2, Nemeth Ch.2
7		Linux System Architecture: Shell, FS	Lecture + Diagrams	Das Ch.2, Nemeth Ch.2

8		Navigation Commands: cd, ls, pwd, mkdir,	Lecture + Demo	Nemeth Ch.2
9		rm, cp, mv	Lab Practice	Nemeth Ch.2
10		Text Editors: vi / vim	Hands-on Lab	Nemeth Ch.3, Coursera link
11		Review & Quiz	Quiz + Discussion	Nemeth Ch.1-3
12	2	PATH variable	Lecture + Demo	Nemeth Ch.4
13		Essential Commands	Lecture	Nemeth Ch.4
14		Essential Commands: man, echo, passwd, uname, who, date	Lecture + Demo	Nemeth Ch.4
15		File Handling: cat, more, wc	Practice Lab	Nemeth Ch.5, Das Ch.3
16		lp, od, tar, gzip	Lab Practice	Nemeth Ch.5
17		File & Directory Management: pwd, cd, mkdir, rmdir	Lecture + Hands-on	Nemeth Ch.5
18		ls, cp, mv, rm	Lecture + Demo	Nemeth Ch.5
19		Security: File Permissions	Lecture + Demo	Nemeth Ch.6
20		Process & Disk: ps, w, ulimit, du, df	Hands-on	Nemeth Ch.6
21		mount, umount	Lab + Discussion	Nemeth Ch.6
22		Networking: arp, ftp, telnet, rlogin	Lecture	Nemeth Ch.7

23		Searching: find, unlink, unmask	Lecture + Practice Lab	Nemeth Ch.7
24		Review & Quiz	Quiz + Clarifications	Nemeth Ch.4-7
25	3	Text Processing Tools: head, tail, sort	Lecture + Demo	Nemeth Ch.8
26		nl, uniq	Lecture+Practice Lab	Nemeth Ch.8
27		grep, egrep, fgrep	Lecture+Demo + Hands-on	Nemeth Ch.8
28		cut, paste, join	Lab Practice	Nemeth Ch.8
29		tee	Lab Exercises	Nemeth Ch.8
30		File Comparison: comm, cmp, diff	Lecture + Practice	Nemeth Ch.8
31		Data Manipulation: tr	Lecture	Nemeth Ch.8
32		awk Intro & Usage	Lecture + Examples	Blum & Bresnahan Ch.1-2
33		sed: Usage and Scripts	Lab	Nemeth Ch.8
34		Backup Utilities: cpio, tar, gzip	Lab Practice	Nemeth Ch.8
35		Review & Mini Quiz	Quiz + Discussion	Nemeth Ch.8
36		Introduction to Shell Scripts	Lecture + Examples	Das Ch.5, Blum & Bresnahan Ch.1

37	4	Writing & Executing Scripts	Lecture + Hands-on	Blum & Bresnahan Ch.2
38		read and Command Line Arguments	Lecture + Lab Practice	Blum & Bresnahan Ch.2
39		Variables in Shell Scripts	Hands-on	Blum & Bresnahan Ch.2
40		Logical and Conditional Operators (if, case)	Lab Practice	Blum & Bresnahan Ch.3
41		Computations with expr, String Handling	Lecture	Blum & Bresnahan Ch.3
42		Loops: while, for	Lecture	Blum & Bresnahan Ch.3
43		Advanced: set, shift, trap	Lecture + Examples	Blum & Bresnahan Ch.4
44		Debugging Shell Scripts	Hands-on Lab	Blum & Bresnahan Ch.4
45		Input Validation and Data Entry	Practice Exercises	Blum & Bresnahan Ch.4
46		Arrays and Functions	Lecture	Blum & Bresnahan Ch.4
47		Review & Mini Project	Group Project / Demo	Nemeth Ch.8, Blum & Bresnahan
48		awk Structure and Execution	Lecture + Examples	Blum & Bresnahan Ch.5
49		Comments, Operators, Fields	Lecture	Blum & Bresnahan Ch.5
50		Built-in Variables	Lab Practice	Blum & Bresnahan Ch.5

51	5	print, printf, Escape Sequences	Demo + Lab	Blum & Bresnahan Ch.5
52		OFS Variable and User Variables	Lecture	Blum & Bresnahan Ch.5
53		Parameter Passing	Lecture	Blum & Bresnahan Ch.5
54		Control Structures: if-else	Lecture + Demo	Blum & Bresnahan Ch.5
55		Loops: while, for	Hands-on + Lab	Blum & Bresnahan Ch.5
56		Arrays in awk	Lab Practice	Blum & Bresnahan Ch.5
57		User-Defined Functions	Lecture	Blum & Bresnahan Ch.5
58		getline Function	Lecture	Blum & Bresnahan Ch.5
59		Regular Expressions in awk	Lecture + Examples	Blum & Bresnahan Ch.5
60		Advanced Use Cases with awk	Lab+ Case study	Blum & Bresnahan Ch.5
61		sed Advanced Usage	Lab + Discussion	Nemeth Ch.8
62		grep + sed Patterns & Applications	Lecture + Practice	Nemeth Ch.8
63		Backup Scripts with tar, cpio, gzip	Lecture + Hands-on	Nemeth Ch.8
64		Data Archiving Automation	Lecture + Discussion Exercise	Nemeth Ch.8

65	5	Integrating awk and sed in Scripts	Hands-on	Blum & Bresnahan
66		Review of awk and sed	Discussion, Q&A	Blum & Bresnahan
67		Practice Session – Full Scripts	Lab Exercise	Blum & Bresnahan
68		Debugging Complex Scripts	Hands-on	Blum & Bresnahan
69		Real-life Scripting Example 1	Group Activity	Blum & Bresnahan
70		Real-life Scripting Example 2	Group Activity	Blum & Bresnahan
71		Real-life Scripting Example 3	Group Activity	Blum & Bresnahan
72		Mini Project Development sample	Team Work	Nemeth, Blum & Bresnahan
73		Presentations and Demos	Student Presentations	Nemeth, Blum & Bresnahan
74		Course Review and Discussion	Open Discussion	Nemeth, Blum & Bresnahan
75	All	Final Assessment and Feedback	Oral Quiz / Feedback Session	Nemeth, Blum & Bresnahan
	All			

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1	
- Quiz Graded Component 1	10 %
- Code completion / Debugging /Command Execution Task	10 %
CIE-2	25 %
- Written Test	
CIE-3	
- Lab Test	15%
- Viva	5 %
- Record	5 %

Semester End Exams (SEE): 30%	
Mode of Exam	Practical
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

PO/CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	2	-	2	-	-	-
CO2	3	2	3	-	-	-	-	-
CO3	3	3	3	2	-	-	-	-
CO4	3	3	2	2	1	-	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Programme Outcome	1. Application of Knowledge Applying Linux commands and shell scripting directly demonstrates core knowledge application in solving real-world computing problems. 2. Critical Thinking & Creativity Debugging scripts and using text-processing tools like grep, sed, and awk requires analytical thinking and creative problem-solving. 3. Digital Transformation The course builds proficiency in using operating system tools, enabling confident and responsible use of digital systems. 4. Research & Innovation Developing automation through scripting encourages exploration and innovative thinking in managing repetitive tasks efficiently. 5. Lifelong Learning Learning Linux and scripting lays the foundation for self-driven technical learning in open-source environments, which constantly evolve.					
Total Marks	70					
CIE components	CO-PO	Excellent (10)	Very Good (7-9)	Good(4-6)	Average(1-3)	Needs Improvement (0)
CIE-1 Graded Component 1 (Quiz) Weightage: 10	CO1,CO2 PO1-PO3, PO4	All answers correct, demonstrating deep understanding of linux and shell Programming fundamentals and concepts with clear explanations	Most answers correct with minor errors, showing good understanding	Some correct answers with partial understanding and notable errors	Few correct answers with significant gaps in understanding	No correct answers or no submission

CIE components	CO-PO	Excellent (10)	Very Good (7-9)	Good (4-6)	Average (1-3)	Needs Improve ment (0)																		
CIE-1 Graded Component 2 (Code completion / Debugging /Command Execution Task) Weightage: 10	CO1,CO2 PO1-PO3	Code is fully correct, optimized, and well-commented. Handles edge cases and found exact logic	Minor mistakes or inefficiencies. The logic is mostly correct. Minor edge cases missed.	Basic logic implemented. Some errors or incomplete handling.	Major logic flaws. Code partially runs but is incomplete or error-prone.	No submission or irrelevant work																		
CIE components	CO-PO	Continuous Internal Evaluation-2 (25 Marks)																						
CIE-2 Mid Sem Examination (Theory) Weightage: 25	CO1, CO2, CO3 PO1-PO4	Mid Sem Examination (Theory) <table><tr><td colspan="3">Mid Sem Assessment Pattern: 25 Marks</td></tr><tr><td>Sl#</td><td colspan="2">Content</td></tr><tr><td colspan="3">Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)</td></tr><tr><td>1</td><td>5 Questions of 1 mark each</td><td>5</td></tr><tr><td colspan="3">Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)</td></tr><tr><td>2</td><td>4 Questions of 5 marks each with 2 or 3 sub divisions possibly</td><td>20</td></tr></table>					Mid Sem Assessment Pattern: 25 Marks			Sl#	Content		Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)			1	5 Questions of 1 mark each	5	Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)			2	4 Questions of 5 marks each with 2 or 3 sub divisions possibly	20
Mid Sem Assessment Pattern: 25 Marks																								
Sl#	Content																							
Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)																								
1	5 Questions of 1 mark each	5																						
Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)																								
2	4 Questions of 5 marks each with 2 or 3 sub divisions possibly	20																						
CIE components	CO-PO	Excellent (10)	Very Good (7-9)	Good (4-6)	Average (1-3)	Needs Improve ment (0)																		
CIE-3 Lab Test Weightage: 15	CO1-CO4 PO1-PO5	Refer Table 1																						
CIE-3 Viva Weightage: 5	CO1-CO4 PO1-PO5																							

CIE-3 Records Weightage: 5		Excellent (5)	Very Good (4)	Good (3)	Average (1-2)	Needs Improvement (0)
		Code is fully correct, optimized, and well-commented. Handles edge cases	Minor mistakes or inefficiencies. The logic is mostly correct. Minor edge cases missed.	Basic logic implemented. Some errors or incomplete handling.	Major logic flaws. Code partially runs but is incomplete or error-prone.	No submission or irrelevant work

Table 1:

Lab Write up Execution Rubrics (15 Marks)

Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor	Marks	Remarks
1	Understanding of problem statements (5 Marks)	Observations	Student exhibits a thorough understanding of the problem statements and applies appropriate data structure and python knowledge to devise a solution. (5 M)	Student has sufficient understanding of the problem statements and applies appropriate data structure and python knowledge to devise a solution. (<5 M and >=1 M)	Student does not have a clear understanding of the problem statements and is unable to apply appropriate data structure and python knowledge to devise a solution. (0 M)		
2	Execution (5 Marks)	Observations	Student demonstrates the execution of the program with appropriate data structure. All test cases are handled with appropriate validations. (5 M)	Student demonstrates the execution of the program with appropriate data structure with only a few test cases handled. (<5 M and >=1 M)	Student has not executed the program or the code fails to demonstrate correct use of data structure. (0 M)		

3	Results and Documentation (5 Marks)	Observations	Documentation with appropriate comments and output is covered in manual. (5 M)	Documentation with only few comments and only few output cases is covered in manual. (<5 M and ≥ 1 M)	Documentation with no comments and no output cases is covered in manual. (0 M)		
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Table 2 : Viva Voice Rubrics (5 Marks)

Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor	Marks	Remarks
1	Conceptual Understanding (5 Marks)	Viva Voce	Explains thoroughly the core concepts and principles of data structure used in corresponding lab as well as python constructs. (5 M)	Adequately explains the core concepts and principles of data structure used in corresponding lab as well as python constructs with some understanding . (<5 M and ≥ 1 M)	Unable to explain the core concepts and principles of data structure used in corresponding lab as well as python constructs. (0 M)		

Signature of the Course Faculty